

Cycle Embedding in the Hypercube with faulty nodes

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Abstract

The hypercube is one of the most versatile and efficient interconnection networks yet discovered for parallel computation. In this paper, we show that a fault-free cycle of length of at least $2^n - 2f$ can be embedded in an n dimensional hypercube with f faulty nodes, where $n \geq 3$ and $1 \leq f \leq 2n - 4$. The best known results thus far can tolerant only $n - 1$ faulty nodes.

Index Terms: fault tolerance, graph embedding, hypercube, cycle, interconnection network, parallel processing.

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1 Introduction

The hypercube is one of the most versatile and efficient interconnection networks (networks for short) yet discovered for parallel computation. It is well suited to both special-purpose and general-purpose tasks, and it can efficiently simulate many other networks of the same size [10]. An *embedding* of one *guest graph* G into another *host graph* H is a one-to-one mapping f from the node set of G to the node set of H [10]. An edge of G corresponds to a path of H under f . Embedding has been the subject of intensive studies with the hypercube as the host graph and various graphs as the guest graph.

Linear arrays and rings, which are two of the most fundamental networks for parallel and distributed computation, are suitable for developing simple algorithms with low communication costs. Many efficient algorithms designed on linear arrays and rings for solving a variety of algebraic problems and graph problems can be found in [1], [10]. They can also be used as control/data flow structures for distributed computation in arbitrary networks. There is an application of longest paths to a practical problem that was encountered in the on-line optimization of a complex Flexible Manufacturing System (see [2]). These motivate the embedding of paths and cycles in networks. Since faults may develop in a practical network, it is important to consider faulty networks.

A *Hamiltonian cycle* in a network W is a cycle that contains every node of W exactly once; thus, W is *Hamiltonian* if there is a Hamiltonian cycle in W [3]. W is *k-link Hamiltonian* if it remains Hamiltonian after removing any k links [6]. Previous results about k -link Hamiltonian networks are as follows. The n -dimensional folded hypercube is $(n - 1)$ -link Hamiltonian [15]. The n -dimensional star graph is $(n - 3)$ -link Hamiltonian [13]. The arrangement graph $A_{n,k}$ is $(k(n - k) - 2)$ -link Hamiltonian [7]. The WK-recursive network of degree d is $(d - 3)$ -link Hamiltonian [4]. A modification of a d -ary undirected de Bruijn graph is $(d - 1)$ -link Hamiltonian [11].

There were fault-tolerant cycle embeddings where the cycle lengths depend on the numbers of faults. For example, a fault-free cycle of length of at least $n! - 2f$ can be embedded in an n -dimensional star graph with f faulty nodes, where $1 \leq f \leq n - 3$ [8]. A fault-free cycle of length of at least $d^n - nf - 1$ can be embedded in a d -ary n -dimensional undirected de Bruijn graph with f faulty nodes, where $1 \leq f \leq d - 1$ [12].

Many results regarding fault-tolerant cycle embedding in a hypercube host graph have been proposed. Latifi et al. showed that the n dimensional hypercube (n -cube) is $(n - 2)$ -link Hamiltonian [9]. Yang et al. showed that a fault-free cycle of length of at least $2^n - 2f$ can be embedded in an n -cube with f faulty nodes, where $1 \leq f \leq n - 2$ [16]. Tseng showed that given an n -cube with $f_e \leq n - 4$ faulty links and $f_v \leq n - 1$ faulty nodes such that $f_e + f_v \leq n - 1$, a cycle of length of at least $2^n - 2f_v$ can be obtained [14].

In this paper, we only consider the problem of embedding a fault-free cycle in an n -cube with faulty nodes. Using an induction-proving scheme, we show that a fault-free cycle of length of at least $2^n - 2f$ can be embedded in an n -cube with f faulty nodes, where $n \geq 3$ and $1 \leq f \leq 2n - 4$, which significantly improve the maximum number of faulty nodes tolerable.

2 Preliminaries

A binary n -cube is an undirected graph with 2^n nodes each labeled with a distinct binary string $b_1b_2 \dots b_n$. Node $b_1 \dots b_i \dots b_n$ and node $b_1 \dots \overline{b_i} \dots b_n$ are joined by an edge along dimension i , where $1 \leq i \leq n$ and $\overline{b_i}$ is the one's complement of b_i . Suppose $X = x_1x_2 \dots x_n$ and $Y = y_1y_2 \dots y_n$. In the rest of the paper, we use $X^{(i)}$ to denote the binary string $x_1 \dots \overline{x_i} \dots x_n$ and use $d_H(X, Y)$ to denote the *Hamming distance* between X and Y , i.e., the number of different bits between X and Y . A path from X to Y is abbreviated to an X - Y path. On the other hand, we say that a node is healthy if it is not faulty.

Conveniently, an n -cube can be represented with $\overbrace{**\dots*}^n = *^n$, where $* \in \{0, 1\}$. Hence, $*^{n-1}1$ and $*^{n-1}0$, which contain the nodes with rightmost bits 1 and 0, respectively, represent two node-disjoint $(n - 1)$ -cubes. The following lemma was shown in [5], which will be used very often.

Lemma 1. [5] Let X and Y be two distinct nodes in an n -cube and $d_H(X, Y) = d$, where $n \geq 1$. There are X - Y paths in the n -cube whose lengths are $d, d + 2, d + 4, \dots, c$, where $c = 2^n - 1$ if d is odd, and $c = 2^n - 2$ if d is even.

3 The Embedding method

Our method is based on a recursive construction. Suppose that we have a cycle C_1 in $*^{n-1}0$ and have a cycle C_2 in $*^{n-1}1$. A basic idea is to combine these two cycles into a longer cycle in $*^n$ (see figure 1). However, we must find a link $U \rightarrow V$ in C_1 such that $U^{(n)} \rightarrow V^{(n)}$ is a link in C_2 . It is possible that we cannot find such link. Hence, we would like to find a link $U \rightarrow V$ in C_1 such that $U^{(n)}$ and $V^{(n)}$ are healthy ($U^{(n)} \rightarrow V^{(n)}$ is not necessarily a link in C_2), then construct another longest $U^{(n)}$ - $V^{(n)}$ path in $*^{n-1}1$ instead of using the original cycle C_2 . The following lemma serves this purpose.

Lemma 2. Let $X \rightarrow Y$ be a link in an n -cube with $f \leq n - 2$ faulty nodes, where $n \geq 2$. There exists a fault-free X - Y path of length of at least $2^n - 2f - 1$.

Proof. We proceed by induction on n . Clearly, the lemma holds for $n = 2, 3$. Assume that it holds for some $k \geq 3$. The situation of $n = k + 1$ is discussed below.

Without loss of generality, assume that $X \rightarrow Y$ belongs to $*^k0$. Let f_0 and f_1 be the numbers of faulty nodes in $*^k0$ and $*^k1$, respectively, where $f_0 + f_1 = f \leq k - 1$. We first consider $f_0 < f$ and $f_1 < f$. By as-

sumption, there exists a fault-free X - Y path of length of at least $2^k - 2f_0 - 1$ in $*^k 0$. Let $U \rightarrow V$ be a link in the fault-free X - Y path such that $U^{(k+1)}$ and $V^{(k+1)}$ are healthy. By assumption, there exists a fault-free $U^{(k+1)}$ - $V^{(k+1)}$ path of length of at least $2^k - 2f_1 - 1$ in $*^k 1$. A fault-free X - Y path of length of at least $(2^k - 2f_0 - 1) - 1 + 2 + (2^k - 2f_1 - 1) = 2^{k+1} - 2f - 1$ in $*^{k+1}$ can be constructed as follows (refer to figure 2).

$$X \Rightarrow U \rightarrow U^{(k+1)} \Rightarrow V^{(k+1)} \rightarrow V \Rightarrow Y,$$

where $U^{(k+1)} \Rightarrow V^{(k+1)}$ is the $U^{(k+1)}$ - $V^{(k+1)}$ path.

Then we consider $f_0 = f$. The discussion for $f_1 = f$ is similar. Let F be a faulty node in $*^k 0$. By assumption (imagining F not faulty), there exists an X - Y path of length of at least $2^k - 2(f - 1) - 1$ in $*^k 0$ that does not contain the other $f - 1$ faulty nodes. We assume that the X - Y path contains F (the discussion is easier when it does not contain F). Let U and V be the two neighbors of F in the X - Y path. We have $d_H(U^{(k+1)}, V^{(k+1)}) = 2$. By Lemma 1, there exists a $U^{(k+1)}$ - $V^{(k+1)}$ path of length $2^k - 2$ in $*^k 1$. Similarly, there exists a fault-free X - Y path of length of at least $(2^k - 2(f - 1) - 1) + 2 - 2 + (2^k - 2) = 2^{k+1} - 2f - 1$ in $*^{k+1}$.

□

Let X and Y be two adjacent nodes in an n -cube. If $X^{(1)}, X^{(2)}, \dots, X^{(n-1)}$ are faulty, clearly, we cannot construct an X - Y path of length $2^n - 2(n - 1) - 1$ in the n -cube. Hence, the f value of Lemma 2 is optimal. However, we may construct a Z - Y path of length $2^n - 2(n - 1) - 1$ in the n -cube, where $Z \neq X$ is a node in the n -cube. Therefore, we can tolerant more faulty nodes in our longest fault-free cycle construction. We will use Lemma 6 to show this result, but first we need Lemmas 3, 4, and 5 to support it.

Lemma 3. Let X and Y be two nodes in an n -cube with an $f \leq 1$ faulty node, where $d_H(X, Y) = 2$ and $n \geq 3$. There exists a fault-free X - Y path of length of at least $2^n - 2f - 2$.

Proof. If $f=0$, then this lemma can be directly obtained from Lemma 1. We assume $f=1$ in the rest of the proof. We proceed by induction on n . Clearly, the lemma holds for $n=3$. Assume that it holds for some $k \geq 3$. The situation of $n=k+1$ is discussed below.

Without loss of generality, assume $X \in {}^k0$ and $Y \in {}^k0$. We first consider the faulty node to be in k0 . By assumption, there exists a fault-free X - Y path of length of at least $2^k - 2f - 2$ in k0 . Let $U \rightarrow V$ be a link in the fault-free X - Y path. By Lemma 1, there exists a $U^{(k+1)}-V^{(k+1)}$ path of length $2^k - 1$ in k1 . A fault-free X - Y path of length of at least $(2^k - 2f - 2) - 1 + 2 + (2^k - 1) = 2^{k+1} - 2f - 2$ in ${}^{k+1}$ can be constructed as shown in figure 2.

Then we consider that the faulty node is in k1 . By Lemma 1, there exists a X - Y path of length $2^k - 2$ in k0 . Let $U \rightarrow V$ be a link in the X - Y path such that $U^{(k+1)}$ and $V^{(k+1)}$ are healthy. By Lemma 2, there exists a fault-free $U^{(k+1)}-V^{(k+1)}$ path of length $2^k - 2f - 1$ in k1 . A fault-free X - Y path of length of at least $(2^k - 2) - 1 + 2(2^k - 2f - 1) = 2^{k+1} - 2f - 2$ in ${}^{k+1}$ can be constructed as shown in figure 2. $\square$

Lemma 4. Let X and Y be two nodes in an n -cube with an $f \leq 1$ faulty node, where $d_H(X, Y) = 3$ and $n \geq 3$. There exists a fault-free X - Y path of length of at least $2^n - 2f - 1$.

Proof. The proof is similar to that of Lemma 3.

Lemma 5. Let e_0 and e_1 be two distinct links in an n -cube with an $f \leq 1$ faulty node, where $n \geq 3$. Suppose e_0 connects nodes X and Y . There exists a fault-free X - Y path of length of at least $2^n - 2f - 1$ in the n -cube that contains e_1 .

Proof. We assume that $f=1$, and the faulty node is F (the proof is easier when $f=0$). We proceed by induction on n . Clearly, the lemma holds for $n=3$. Assume that it holds for some $k \geq 3$. The situation of $n=k+1$ is discussed below.

Since $k + 1 \geq 4$, there exists a bit position, say j , such that each of e_0 and e_1 belongs to $*^{j-1}0*^{k+1-j}$ or $*^{j-1}1*^{k+1-j}$. Without loss of generality, let $j = k + 1$. We assume $e_0 \in *^k0$. The discussion for $e_0 \in *^k1$ is similar.

When $e_1 \in *^k0$ and $F \in *^k0$, by assumption, there exists a fault-free X - Y path of length of at least $2^k - 2f - 1$ in $*^k0$ that contains e_1 . Let $U \rightarrow V$ be a link in the X - Y path different from e_1 . By Lemma 1, there exists a $U^{(k+1)}$ - $V^{(k+1)}$ path of length $2^k - 1$ in $*^k1$. A fault-free X - Y path of length of at least $(2^k - 2f - 1) - 1 + 2 + (2^k - 1) = 2^{k+1} - 2f - 1$ in $*^{k+1}$ that contains e_1 can be constructed as shown in figure 2.

When $e_1 \in *^k0$ and $F \in *^k1$, by assumption, there exists a fault-free X - Y path of length of at least $2^k - 1$ in $*^k0$ that contains e_1 . Let $U \rightarrow V$ be a link in the X - Y path different from e_1 such that $F \notin \{U^{(k+1)}, V^{(k+1)}\}$. By Lemma 2, there exists a fault-free $U^{(k+1)}$ - $V^{(k+1)}$ path of length $2^k - 2f - 1$ in $*^k1$. A fault-free X - Y path of length of at least $(2^k - 1) - 1 + 2 + (2^k - 2f - 1) = 2^{k+1} - 2f - 1$ in $*^{k+1}$ that contains e_1 can be constructed as shown in figure 2.

The cases of $e_1 \in *^k1$ and $F \in *^k1$, and $e_1 \in *^k1$ and $F \in *^k0$ can be proved with similar discussions.

□

Lemma 6. Let X and Y be two adjacent nodes in an n -cube with $n - 1$ faulty nodes such that besides $Y(X)$, at least one other healthy node is adjacent to $X(Y)$, where $n \geq 3$. There exists a fault-free X - Y path of length of at least $2^n - 2(n - 1) - 1$.

Proof. We proceed by induction on n . Clearly, the lemma holds for $n = 3$. Assume that it holds for some $k \geq 3$. The situation of $n = k + 1$ is discussed below.

Since $k + 1 \geq 4$, there exists a bit position, say j , such that (1) the number of faulty nodes in $*^{j-1}0*^{k+1-j} (*^{j-1}1*^{k+1-j})$ is at least 1, and (2) both X and Y belong to $*^{j-1}0*^{k+1-j}$ or $*^{j-1}1*^{k+1-j}$. Let f_0 and f_1 be the numbers of faulty nodes in $*^{j-1}0*^{k+1-j}$ and $*^{j-1}1*^{k+1-j}$, respectively, where $f_0 + f_1 = k$. Without loss of generality, let $j = k + 1$, and X and Y belong to $*^k0$. Four cases are considered:

Case 1. $f_0 = k - 1$ (i.e., $f_1 = 1$). Three cases are further considered:

Case 1.1. All nodes in $*^k0$ (except Y) adjacent to X are faulty. Hence, all nodes in $*^k0$ not adjacent to X are healthy. Let F be one faulty node in $*^k0$. By Lemma 2 (imagining F not faulty), there exists an X - Y path of length of at least $2^k - 2(f_0 - 1) - 1$ in $*^k0$ that does not contain the other $f_0 - 1$ faulty nodes. Let $U \neq X$ be another neighbor of F in the X - Y path.

First we consider that $U^{(k+1)}$ is healthy. We have $d_H(X^{(k+1)}, U^{(k+1)}) = 2$. By Lemma 3, there exists a fault-free $X^{(k+1)}$ - $U^{(k+1)}$ path of length of at least $2^k - 2f_1 - 2$. A fault-free X - Y path of length of at least $(2^k - 2(f_0 - 1) - 1) - 2 + 2 + (2^k - 2f_1 - 2) = 2^{k+1} - 2k - 1$ can be constructed as shown in figure 3.

Then we consider that $U^{(k+1)}$ is faulty. Let $S \neq F$ be another neighbor of U in X - Y path. We have $d_H(S^{(k+1)}, X^{(k+1)}) = 3$. By Lemma 4, there exists a fault-free $S^{(k+1)}$ - $X^{(k+1)}$ path of length of at least $2^k - 2f_1 - 1$. A fault-free X - Y path of length of at least $(2^k - 2(f_0 - 1) - 1) - 3 + 2 + (2^k - 2f_1 - 1) = 2^{k+1} - 2k - 1$ can be constructed as shown in figure 4.

Case 1.2. All nodes in $*^k0$ (except X) adjacent to Y are faulty. Proof is similar to that of Case 1.1.

Case 1.3. Besides Y (X), at least one other healthy node in $*^k0$ is adjacent to X (Y). By assumption, there exists a fault-free X - Y path of length of at least $2^k - 2f_0 - 1$. Let $U \rightarrow V$ be a link in the X - Y path such that $U^{(k+1)}$ and $V^{(k+1)}$ are healthy. By Lemma 2, there exists a fault-free $U^{(k+1)}$ - $V^{(k+1)}$ path of length 2^k

$-2f_1 - 1$ in $*^k 1$. A fault-free cycle of length of at least $(2^k - 2f_0 - 1) - 1 + 2 + (2^k - 2f_1 - 1) = 2^{k+1} - 2k - 1$ in $*^{k+1}$ can be constructed as shown in figure 2.

Case 2. $2 \leq f_0 \leq k - 2$, (i.e., $2 \leq f_1 \leq k - 2$). By Lemma 2, there exists a fault-free X - Y path of length of at least $2^k - 2f_0 - 1$ in $*^k 0$. Let $U \rightarrow V$ be a link in the X - Y path such that $U^{(k+1)}$ and $V^{(k+1)}$ are healthy. Also, by Lemma 2, there exists a fault-free $U^{(k+1)}$ - $V^{(k+1)}$ path of length $2^k - 2f_1 - 1$ in $*^k 1$. A fault-free X - Y path of length of at least $(2^k - 2f_0 - 1) - 1 + 2 + (2^k - 2f_1 - 1) = 2^{k+1} - 2k - 1$ in $*^{k+1}$ can be constructed as shown in figure 2.

Case 3. $f_0 = 1$ (i.e., $f_1 = k - 1$). Let $U^{(k+1)} \rightarrow V^{(k+1)}$ be a link in $*^k 1$ such that (1) besides $U^{(k+1)}$ ($V^{(k+1)}$), at least one other healthy node is adjacent to $V^{(k+1)}$ ($U^{(k+1)}$) in $*^k 1$, and (2) U and V are healthy. By Lemma 5, there exists a fault-free X - Y path of length of at least $2^k - 2f_0 - 1$ in $*^k 0$ that contains $U \rightarrow V$. In addition, by assumption, there exists a fault-free $U^{(k+1)}$ - $V^{(k+1)}$ path of length of at least $2^n - 2f_1 - 1$ in $*^k 1$. A fault-free X - Y path of length of at least $(2^k - 2f_0 - 1) - 1 + 2 + (2^k - 2f_1 - 1) = 2^{k+1} - 2k - 1$ in $*^{k+1}$ can be constructed as shown in figure 2. □

The following theorem concludes our main result in this paper.

Theorem 1. There exists a fault-free cycle of length of at least $2^n - 2f$ in an n -cube with f faulty nodes, where $1 \leq f \leq 2n - 4$ and $n \geq 3$.

Proof. In [14], Tseng showed that fault-free cycles of lengths of at least $2^n - 2f$ can be embedded in an n -cube with f faulty nodes, where $1 \leq f \leq n - 1$. In following discussion, we assume $f \geq n - 1$. We proceed by induction on n . Clearly, the lemma holds for $n = 3$. Assume that it holds for some $k \geq 3$. The situation of $n = k + 1$ is discussed below.

Since $f \geq k \geq 3$, there exists a bit position, say j , such that the number of faulty nodes in $*^{j-1}0*^{k+1-j}$ ($*^{j-1}1*^{k+1-j}$) is at least 1. Let f_0 and f_1 be the numbers of faulty nodes in $*^{j-1}0*^{k+1-j}$ and $*^{j-1}1*^{k+1-j}$, respectively, where $f_0 + f_1 = f$. Without loss of generality, let $j = k + 1$ and $f_0 \geq f_1 \geq 1$. Four cases are considered:

Case 1. $f_0 = 2k - 3$. (i.e., $f_1 = 1$). Let F be a faulty node in $*^k0$. By assumption (imagining F not faulty), there exists a cycle C of length of at least $2^k - 2(f_0 - 1)$ in $*^k0$ that does not contain the other faulty nodes. We assume that C contains F (the discussion is easier when it does not contain F). Let U and V be the two neighbors of F in C .

First we consider that both $U^{(k+1)}$ and $V^{(k+1)}$ are healthy. We have $d_H(U^{(k+1)}, V^{(k+1)}) = 2$. By Lemma 3, there exists a fault-free $U^{(k+1)}$ - $V^{(k+1)}$ path of length of at least $2^k - 2f_1 - 2$. A fault-free cycle of length of at least $(2^k - 2(f_0 - 1)) - 2 + 2 + (2^k - 2f_1 - 2) = 2^{k+1} - 2f$ in $*^{k+1}$ can be constructed as shown in figure 5.

Then we consider that $U^{(k+1)}$ is faulty (the discussion of $V^{(k+1)}$ being faulty is the same). Let $S \neq F$ be the neighbor of U in cycle C . We have $d_H(S^{(k+1)}, V^{(k+1)}) = 1$ or 3. By Lemmas 2 and 4, there exists a fault-free $S^{(k+1)}$ - $V^{(k+1)}$ path of length of at least $2^k - 2f_1 - 1$. A fault-free cycle of length of at least $(2^k - 2(f_0 - 1)) - 3 + 2 + (2^k - 2f_1 - 1) = 2^{k+1} - 2f$ in $*^{k+1}$ can be constructed as shown in figure 6.

Case 2. $k \leq f_0 \leq 2k - 4$ (i.e., $1 \leq f_1 \leq k - 2$). We have $k \geq 4$. By assumption, there exists a fault-free cycle of length of at least $2^k - 2f_0$ in $*^k0$. Let $U \rightarrow V$ be a link in $*^k0$ such that $U^{(k+1)}$ and $V^{(k+1)}$ are healthy. In addition, by Lemma 2, there exists a fault-free $U^{(k+1)}$ - $V^{(k+1)}$ path of length of at least $2^k - 2f_1 - 1$ in $*^k1$. A fault-free cycle of length of at least $(2^k - 2f_0) - 1 + 2 + (2^k - 2f_1 - 1) = 2^{k+1} - 2f$ in $*^{k+1}$ can be constructed as shown in figure 7.

Case 3. $f_0 = k - 1$ (i.e., $1 \leq f_1 \leq k - 1$). When $f_1 \leq k - 2$, the discussion is the same as in Case 2. We assume that $f_1 = k - 1$ in the following discussion. By assumption, there exists a fault-free cycle of length of at least $2^k - 2f_0$ in $*^k 0$. Let $U \rightarrow V$ be a link in $*^k 0$ such that (1) $U^{(k+1)}$ and $V^{(k+1)}$ are healthy, and (2) besides $U^{(k+1)}$ ($V^{(k+1)}$), at least one other healthy node is adjacent to $V^{(k+1)}$ ($U^{(k+1)}$). (We can find such a link except for in the situation shown in figure 8, which also can construct a fault-free cycle of length $2^{k+1} - 2f = 8$.) In addition, by Lemma 6, there exists a fault-free $U^{(k+1)}$ - $V^{(k+1)}$ path of length of at least $2^k - 2f_1 - 1$ in $*^k 1$. A fault-free cycle of length of at least $(2^k - 2f_0) - 1 + 2 + (2^k - 2f_1 - 1) = 2^{k+1} - 2f$ in $*^{k+1}$ can be constructed as shown in figure 7. $\square$

4 Discussion and conclusion

Induction proving is a simple and powerful scheme particularly suited to recursive networks. Using an induction-proving scheme, we have showed that the fault-free cycles of lengths of at least $2^n - 2f$ can be embedded in an n -cube with f faulty nodes, where $n \geq 3$ and $1 \leq f \leq 2n - 4$. It is not difficult to design a recursive algorithm according to our proof.

As compared to [14] and [16], this paper significantly improves the maximum number of faulty nodes tolerable. An interesting question is whether we can do better. Figure 9 shows that we cannot embed a fault-free cycle of length of at least $2^4 - 2 \times 5 = 6$ in a 4-cube with 5 faulty nodes. However, it is not easy to prove that a fault-free cycle of length of at least $2^n - 2f$ cannot be embedded in an n -cube with f faulty nodes, where $n \geq 5$ and $f \geq 2n - 3$.

Suppose the minimum degree of a network W is d . The maximum number of faulty nodes tolerable is $d - 2$ in most previous works (see [8], [12]). This paper reveals that we may increase the maximum number of faulty nodes tolerable from $d - 2$ to $2(d - 2)$. To embed fault-free cycles in the star graph with more

faulty nodes is one of our further projects.

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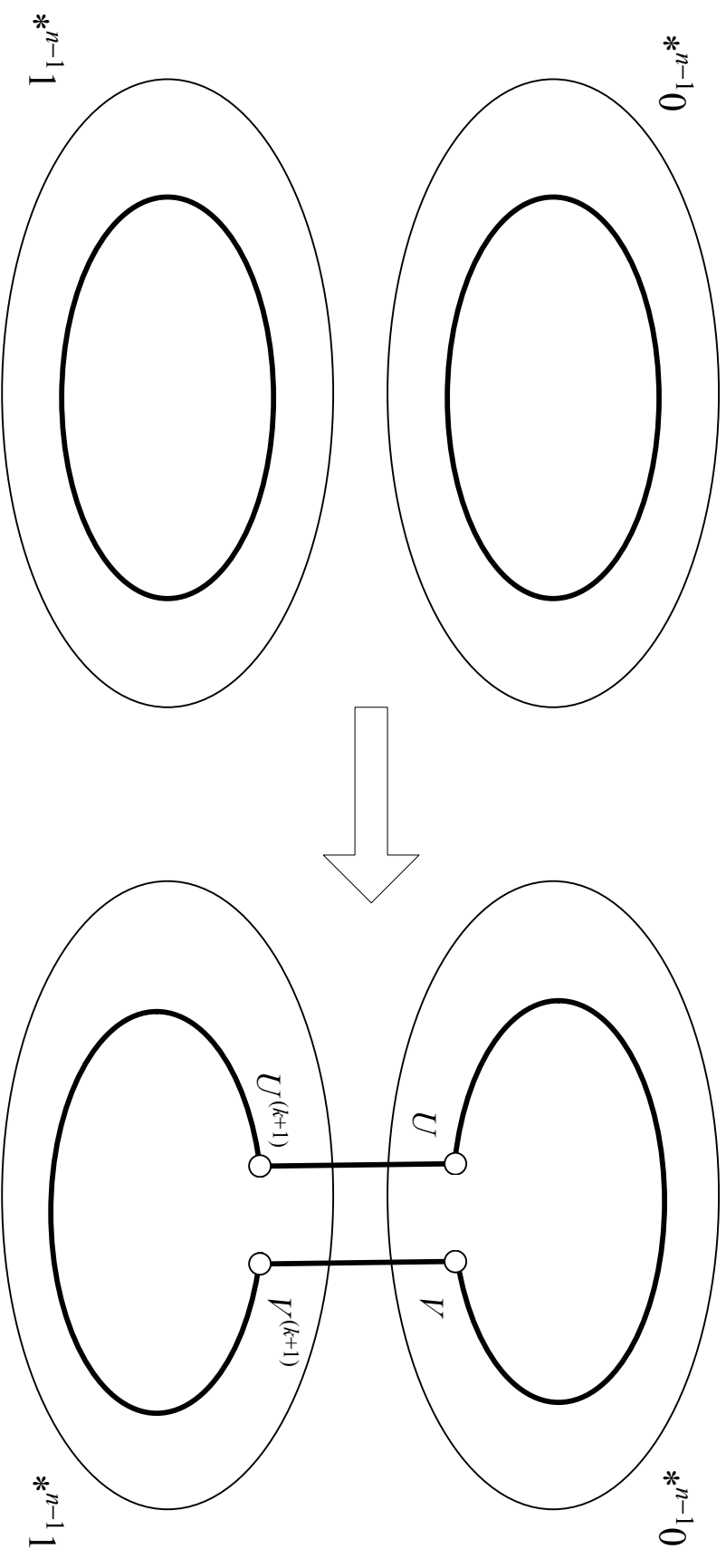


Figure 1. A basic representation of embedding a cycle in $*^n$ from $*^{n-1}0$ and $*^{n-1}1$.

← a path

← a link

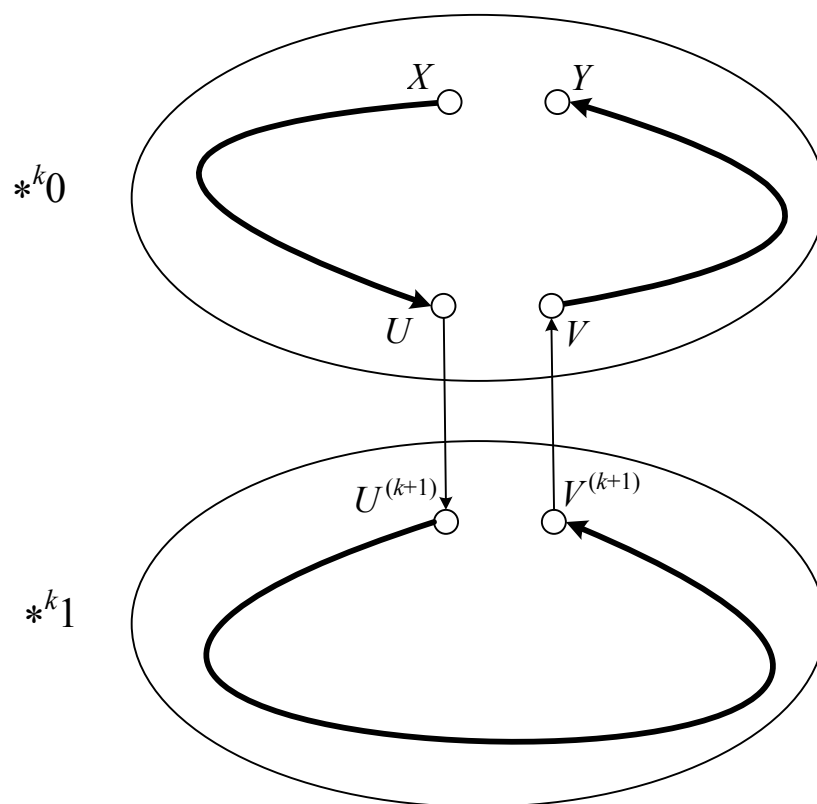


Figure 2. The construction of a fault-free X - Y path in $*^{k+1}$.

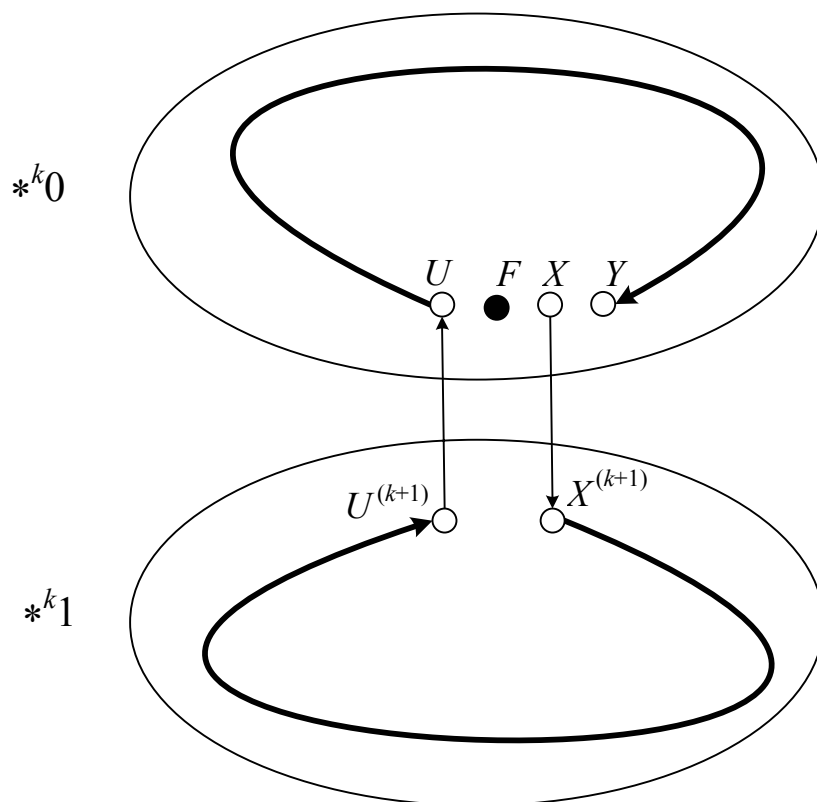


Figure 3. The construction of a fault-free X - Y path in $*^{k+1}$ when $U^{(k+1)}$ is healthy.

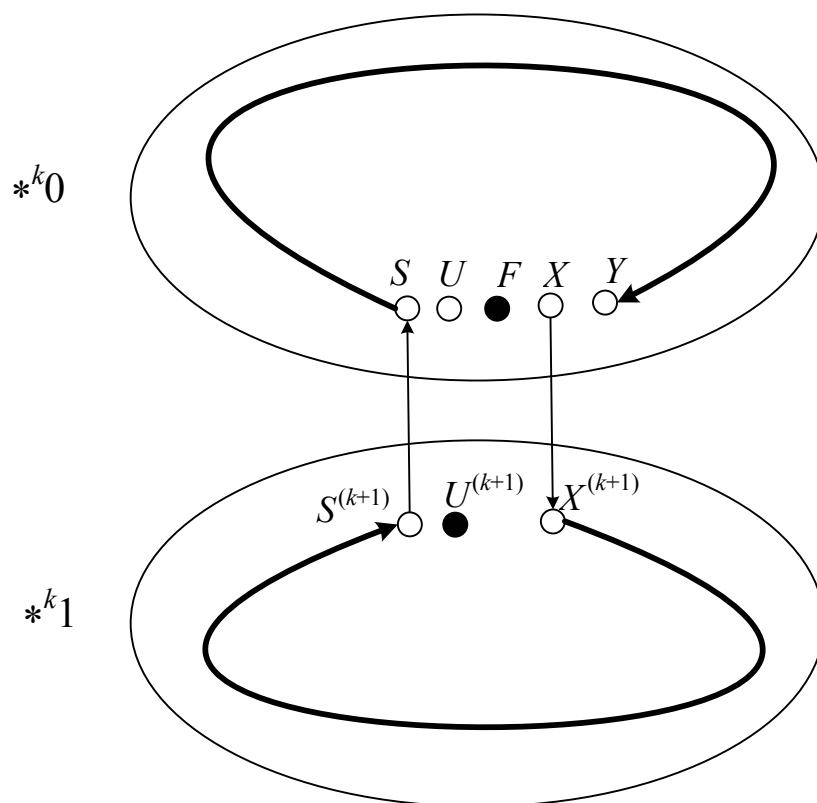


Figure 4. The construction of a fault-free X - Y path in $*^{k+1}$ when $U^{(k+1)}$ is faulty.

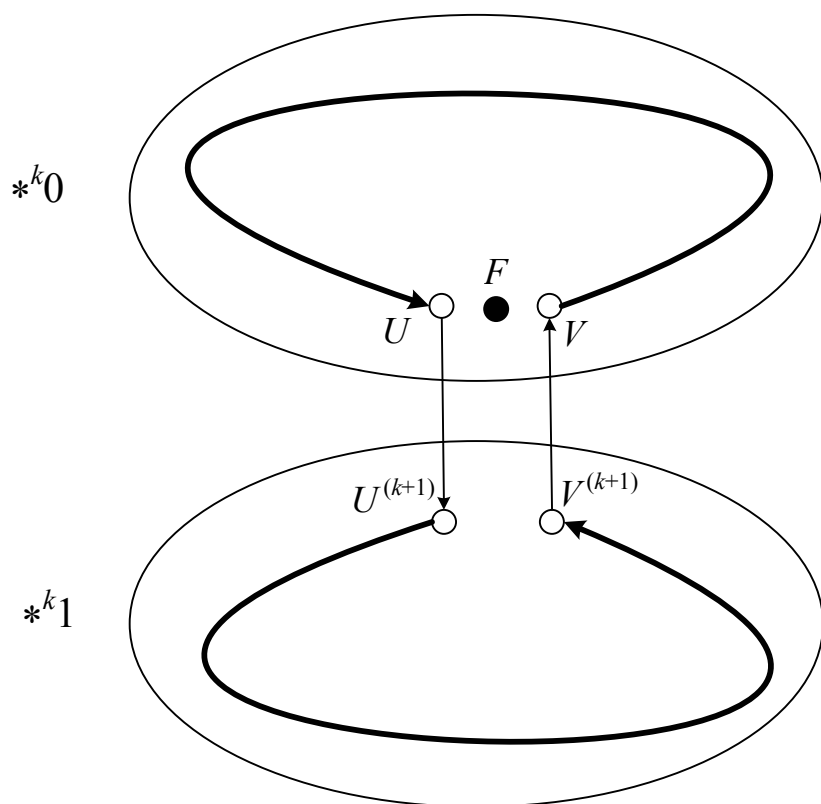


Figure 5. The construction of a fault-free cycle in $*^{k+1}$ when both $U^{(k+1)}$ and $V^{(k+1)}$ are healthy.

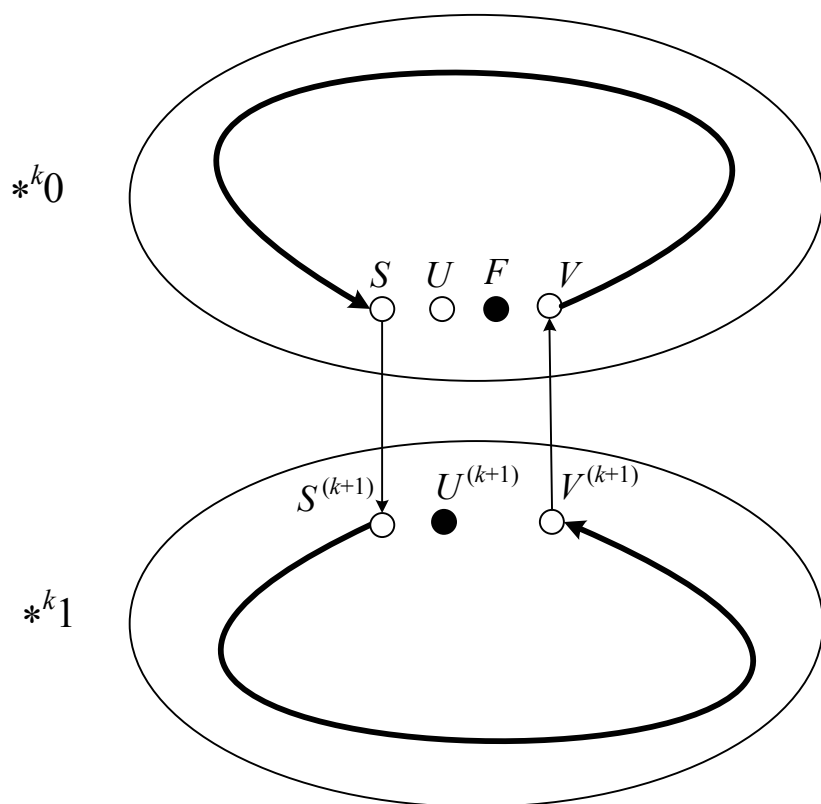


Figure 6. The construction of a fault-free cycle in $*^{k+1}$ when $U^{(k+1)}$ is faulty.

← a path

← a link

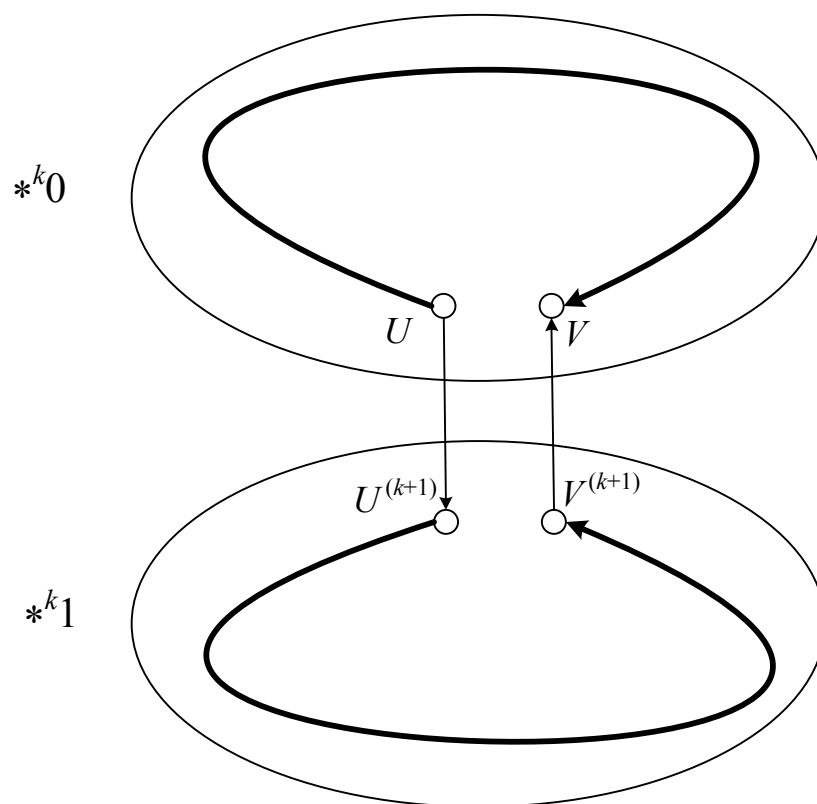


Figure 7. The construction of a fault-free cycle in $*^{k+1}$.

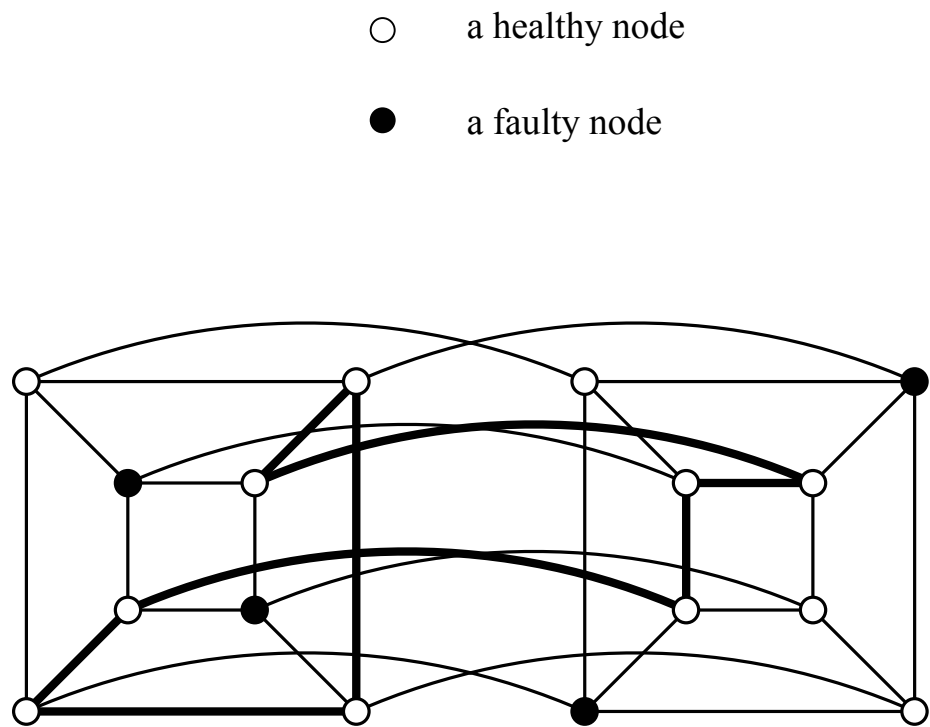


Figure 8. A fault-free cycle of length 8 in the 4-cube with 4 faulty nodes.

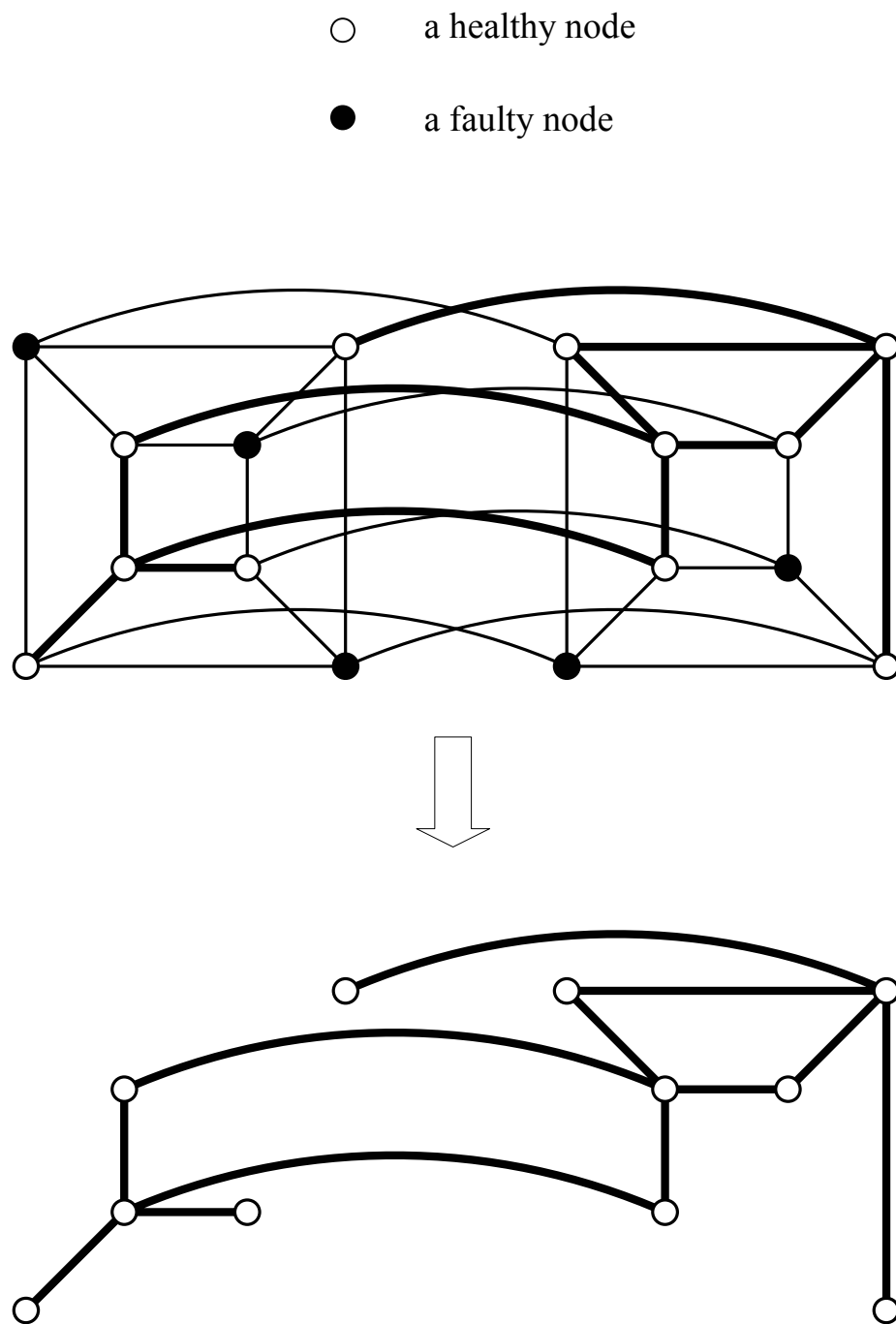


Figure 9. A 4-cube with 5 faulty nodes and the graph after removing the faulty nodes.